Cross-Currency Dynamics in Cryptocurrencies under Stablecoin Regulation

Abstract

Stablecoins are the main quote currency on many crypto exchanges, so their funding and redemption risk can distort the price of the same asset across quote currencies and venues. We study Bitcoin (BTC) quoted in USD, USDT, and USDC using 1-minute OHLCV data from March 1–21, 2023, spanning the Silicon Valley Bank (SVB) shock and the USDC de-peg. We separate (i) *unadjusted dispersion* D_t and (ii) *adjusted parity residual* B_t that marks stablecoins back to USD, estimate mean reversion via an Ornstein–Uhlenbeck (OU) model, and evaluate price discovery on log-price levels using Johansen cointegration and VECM adjustment coefficients (with BH/FDR-adjusted Granger tests as secondary evidence). We find (i) the USDC de-peg produces very large unadjusted dispersion (Kraken crisis mean $D_t \approx +319.97$ bps), while the adjusted residual is much smaller ($B_t \approx +5.27$ bps) and remains fast mean-reverting (USDC crisis half-life 0.61 minutes at 1-minute sampling; robustness 0.54–1.10 minutes), (ii) USDC stress is far larger than USDT, and (iii) the Kraken BTC/USD–BTC/USDT channel is cointegrated with BTC/USD adjusting less to disequilibrium, while the BTC/USD–BTC/USDC channel does not reject no-cointegration under our conservative no-forward-fill BIC specification, so USDC "follows fiat" is treated as supportive rather than primary evidence. We discuss how stablecoin regulation (e.g., the 2025 GENIUS Act) targets the banking-layer fragility that drives these dislocations.

1 Context and Motivation

1.1 Why stablecoins matter for pricing

A stablecoin is a token designed to track a fiat currency (typically USD). In crypto spot markets, stablecoins function like a digital cash account: traders use them to quote prices, move liquidity, and fund positions across exchanges. Because stablecoins are not identical to bank deposits (redeemability, custody, and transparency can differ), their prices can temporarily deviate from $1 and introduce a *stablecoin funding risk premium*. That premium can show up as different BTC prices across quote currencies (BTC/USD vs. BTC/USDC vs. BTC/USDT), even when the underlying asset is the same.

1.2 Regulatory backdrop: the GENIUS Act

In 2025, the U.S. enacted the Guiding and Establishing National Innovation for U.S. Stablecoins Act (the "GENIUS Act"), establishing a federal framework for dollar-pegged stablecoins. The core economic implication is straightforward: stricter reserve backing, transparency, and banking oversight

should reduce uncertainty about redemption and reserves, which should reduce stablecoin-specific risk premia and tighten cross-currency parity.

1.3 SVB and the USDC de-peg as a stress test

To understand how stablecoin stress affects cross-currency pricing, we analyze March 1–21, 2023 (UTC), which includes the SVB failure. When SVB collapsed on March 10, approximately $3.3B of Circle's USDC cash reserves became temporarily inaccessible, and USDC de-pegged (trading as low as $0.87 on March 11–12). This episode is a clean, high-frequency stress test of (i) cross-currency basis behavior, (ii) liquidity fragmentation, and (iii) price discovery across exchanges and quote currencies.

2 Data and Methodology

2.1 Data sources

We collect 1-minute OHLCV candles for BTC quoted in USD, USDT, and USDC from March 1, 2023, 00:00:00 UTC to March 21, 2023, 23:59:00 UTC (30,240 minutes per pair expected). We also download BTC/EUR pairs for completeness, but EUR is not used in the main empirical results. We use three centralized exchanges with different market structure and regulatory exposure:

- **Binance:** BTC/USDT, BTC/USDC.
- **Coinbase:** BTC/USD, BTC/USDT, USDT/USD.
- **Kraken:** BTC/USD, BTC/USDT, BTC/USDC, USDT/USD, USDC/USD.

For $P_{Q/\mathrm{USD},t}$ in $B_{Q,t}$, we use venue-specific stablecoin/USD legs (Kraken USDC/USD and USDT/USD for Kraken channels, and Coinbase USDT/USD for Coinbase channels). Because of historical constraints, Kraken candles are reconstructed by aggregating paginated trade data into 1-minute OHLCV bars.

2.2 Preprocessing and alignment

All series are mapped to a unified 1-minute UTC grid. Missing observations are limited (Binance near 100%, Coinbase/Kraken $> 98\%$). For short micro-gaps (up to 5 minutes), we forward-fill close prices to maintain continuity; larger gaps are flagged to avoid artificial volatility.

Since we do not rely on full order book snapshots, we proxy transaction frictions and depth using the intra-candle range:

$$\text{Intra-minute Range (bps)} = \left(\frac{\text{High}_t - \text{Low}_t}{\text{Close}_t}\right) \times 10000, \tag{1}$$

where "bps" means basis points (100 bps $= 1\%$). This is *not* a quoted bid–ask spread. It is a coarse range proxy that mechanically loads on volatility and microstructure noise, so we interpret it cautiously as a friction indicator.

2.3 Core quantitative objects

2.3.1 Unadjusted dispersion vs. adjusted parity residual

For stablecoin quote currency $Q \in \{\text{USDC}, \text{USDT}\}$, we track two related objects.

First, the *unadjusted dispersion*:

$$D_{Q,t} = \big(\ln(P_{\text{BTC}/Q,t}) - \ln(P_{\text{BTC/USD},t})\big) \times 10000. \tag{2}$$

Second, the *adjusted parity residual* (the paper's original "basis" object):

$$B_{Q,t} = \big(\ln(P_{\text{BTC}/Q,t} \times P_{Q/\text{USD},t}) - \ln(P_{\text{BTC/USD},t})\big) \times 10000. \tag{3}$$

In implementation, $P_{Q/\text{USD},t}$ is venue-specific for each channel (e.g., Kraken $B_{Q,t}$ uses Kraken stablecoin/USD).

These satisfy the accounting identity

$$B_{Q,t} = D_{Q,t} + 10000 \cdot \ln(P_{Q/\text{USD},t}). \tag{4}$$

So large stablecoin de-pegs mechanically move $D_{Q,t}$, while $B_{Q,t}$ captures the residual after marking the stablecoin leg back to USD.

We compute these objects *intra-exchange* (same venue) to isolate stablecoin effects from venue differences; for spatial arbitrage we separately study cross-exchange log-price differentials.

2.3.2 Mean reversion via Ornstein–Uhlenbeck

We model the adjusted residual B_t (denoted X_t below) as an OU process:

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t, \tag{5}$$

and test stationarity with Augmented Dickey–Fuller (ADF) tests. With very large 1-minute samples, ADF rejection is expected and is used here as a basic stationarity check rather than a tradability guarantee. In discrete time, we estimate the error-correction form

$$\Delta X_t = a + \beta X_{t-1} + \varepsilon_t, \tag{6}$$

which implies a levels AR(1):

$$X_t = c + \rho X_{t-1} + \varepsilon_t, \quad \rho = 1 + \beta. \tag{7}$$

For sampling interval Δt (in minutes), we use the exact half-life mapping

$$\tau_{1/2} = \frac{\ln(2)\,\Delta t}{-\ln(\rho)}, \quad 0 < \rho < 1. \tag{8}$$

If $\rho \le 0$ or $\rho \ge 1$, the mean-reversion half-life is undefined and we report NaN. Because 1-minute data can contain microstructure noise, we also report robustness at 5-minute sampling and with forward-filled minutes removed.

2.3.3 Price discovery in levels: Johansen + VECM (primary)

For the two primary channels (Kraken BTC/USD vs. Kraken BTC/USDC and Kraken BTC/USD vs. Kraken BTC/USDT), we use synchronized 1-minute *log-price levels* and drop any minute where either leg is forward-filled. We test for cointegration with Johansen trace tests (constant restricted to the cointegration relation), and choose lag order via BIC on the levels VAR (AIC fallback). If p is the selected VAR lag in levels, we use $k_\Delta = p - 1$ in the VECM.

When rank $= 1$, we estimate:

$$\Delta \mathbf{p}_t = \alpha \beta^\top \mathbf{p}_{t-1} + \sum_{i=1}^{k_\Delta} \Gamma_i \Delta \mathbf{p}_{t-i} + \mathbf{u}_t. \tag{9}$$

The market with smaller $|\alpha|$ adjusts less to disequilibrium and is interpreted as the relative leader in the common efficient price.

For two markets, we also report Gonzalo–Granger component metrics from $\alpha = (\alpha_1, \alpha_2)^\top$:

$$\mathrm{CS}_1 = \frac{\alpha_2}{\alpha_2 - \alpha_1}, \qquad \mathrm{CS}_2 = \frac{-\alpha_1}{\alpha_2 - \alpha_1}. \tag{10}$$

These metrics can be outside $[0, 1]$ in finite samples, so we treat them as supplementary diagnostics. We check robustness by re-estimating at $k_\Delta \pm 1$.

2.3.4 Granger causality (secondary/supportive)

We retain Granger tests on returns as supportive evidence only, not as the primary discovery metric. Across the full Granger test family, we apply Benjamini–Hochberg false discovery rate (BH/FDR) adjustment. Because 1-minute bars can be asynchronous across venues and the sample is very large, statistically significant Granger p-values may overstate economic lead–lag strength; we therefore do not interpret them as structural causality.

3 Empirical Results: Pricing, Liquidity, and Discovery

3.1 Intra-exchange dispersion vs. adjusted residual

Before SVB (March 1–9), both objects are near zero on Kraken. During the crisis (March 10–13), the *unadjusted* USDC dispersion jumps to a mean of $+319.97$ bps with $\sigma = 322.87$ bps, while the *adjusted* USDC residual is much smaller (mean $+5.27$ bps, $\sigma = 21.21$ bps). For USDT on Kraken, crisis means are -57.18 bps for D_t and $+0.68$ bps for B_t (Table 1).

Figure 1: Panel A: intra-exchange adjusted parity residuals (B_t) on Kraken and Coinbase. Panel B: USDC/USDT relative adjusted residual on Binance (no direct fiat exposure).

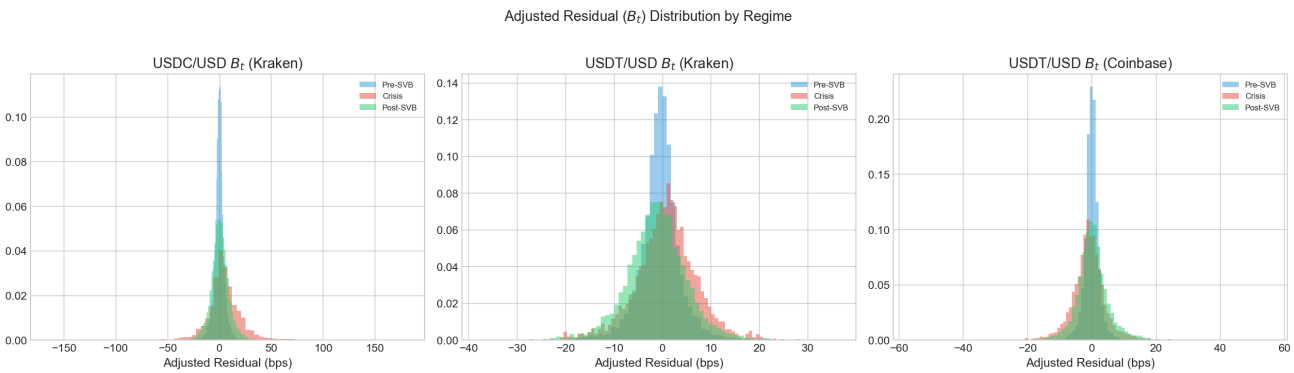

Figure 2: Adjusted residual (B_t) distributions by regime for USDC/USD (Kraken), USDT/USD (Kraken), and USDT/USD (Coinbase).

Table 1: Regime Statistics for Unadjusted Dispersion (D_t) and Adjusted Residual (B_t), Kraken

Regime	Series	Mean (bps)	Std (bps)	Mean Abs (bps)	N
Pre-SVB	USDC Kraken D_t (Unadjusted)	0.12	4.95	3.49	11615
Pre-SVB	USDC Kraken B_t (Adjusted)	-0.32	5.00	3.53	11197
Pre-SVB	USDT Kraken D_t (Unadjusted)	-0.54	4.49	3.17	12618
Pre-SVB	USDT Kraken B_t (Adjusted)	-0.60	4.33	3.02	12618
Crisis	USDC Kraken D_t (Unadjusted)	319.97	322.87	321.17	4311
Crisis	USDC Kraken B_t (Adjusted)	5.27	21.21	13.37	4283
Crisis	USDT Kraken D_t (Unadjusted)	-57.18	45.10	58.57	4312
Crisis	USDT Kraken B_t (Adjusted)	0.68	6.57	4.94	4312
Post-SVB	USDC Kraken D_t (Unadjusted)	11.22	21.95	14.27	12514
Post-SVB	USDC Kraken B_t (Adjusted)	0.45	9.06	6.68	12419
Post-SVB	USDT Kraken D_t (Unadjusted)	-30.26	13.29	30.28	12837
Post-SVB	USDT Kraken B_t (Adjusted)	-1.23	6.47	4.92	12837

Figure 3: SVB window (March 10–13): unadjusted dispersion D_t vs. adjusted residual B_t on Kraken. The USDC de-peg dominates D_t, while B_t remains far smaller after marking the stablecoin leg to USD.

Despite large *unadjusted* dislocations, arbitrage in the adjusted residual remains fast: for B_t series, ADF rejects a unit root across regimes ($p < 0.001$ in the main adjusted-residual channels; with large N, we treat this as a baseline stationarity screen rather than a tradability guarantee). Using the corrected exact mapping, the USDC adjusted residual on Kraken has half-life 0.98 minutes pre-SVB, 0.61 minutes in crisis, and 0.79 minutes post-SVB (1-minute, all observations). Robustness checks (Table 2) show crisis USDC half-life between 0.54 and 1.10 minutes depending on sampling and forward-fill filtering; thus fast mean reversion is robust, while the claim that crisis half-life strictly falls

Table 2: Half-Life Robustness for Adjusted Residuals (B_t): 1m vs 5m and All vs No-Forward-Fill. Some no-FF cells have small n and may be unstable; NaN indicates the estimated rho is outside (0,1). The no-FF sample can be small because stablecoin/USD legs are sparse at 1-minute frequency and many minutes are carried-forward closes.

Series	Regime	Freq	FF Filter	rho	Half-Life (min)	N	Warning
USDC/USD B_t (Kraken)	Pre-SVB	1m	all	0.4944	0.9841	11196	—
USDC/USD B_t (Kraken)	Pre-SVB	1m	no-ff	0.1397	0.3521	1166	—
USDC/USD B_t (Kraken)	Pre-SVB	5m	all	0.0722	1.3189	2478	—
USDC/USD B_t (Kraken)	Pre-SVB	5m	no-ff	-0.1311	NaN	208	rho invalid
USDT/USD B_t (Kraken)	Pre-SVB	1m	all	0.4543	0.8785	12617	—
USDT/USD B_t (Kraken)	Pre-SVB	1m	no-ff	0.2360	0.4800	4619	—
USDT/USD B_t (Kraken)	Pre-SVB	5m	all	0.0998	1.5037	2583	—
USDT/USD B_t (Kraken)	Pre-SVB	5m	no-ff	0.1433	1.7836	863	—
USDC/USD B_t (Kraken)	Crisis	1m	all	0.3203	0.6088	4282	—
USDC/USD B_t (Kraken)	Crisis	1m	no-ff	0.2738	0.5351	2381	—
USDC/USD B_t (Kraken)	Crisis	5m	all	0.0424	1.0968	861	—
USDC/USD B_t (Kraken)	Crisis	5m	no-ff	0.0123	0.7885	470	—
USDT/USD B_t (Kraken)	Crisis	1m	all	0.4877	0.9654	4311	—
USDT/USD B_t (Kraken)	Crisis	1m	no-ff	0.4692	0.9160	2720	—
USDT/USD B_t (Kraken)	Crisis	5m	all	0.2927	2.8209	863	—
USDT/USD B_t (Kraken)	Crisis	5m	no-ff	0.3224	3.0613	525	—
USDC/USD B_t (Kraken)	Post-SVB	1m	all	0.4176	0.7937	12418	—
USDC/USD B_t (Kraken)	Post-SVB	1m	no-ff	0.2029	0.4345	3870	—
USDC/USD B_t (Kraken)	Post-SVB	5m	all	0.0734	1.3270	2559	—
USDC/USD B_t (Kraken)	Post-SVB	5m	no-ff	0.1058	1.5427	720	—
USDT/USD B_t (Kraken)	Post-SVB	1m	all	0.2955	0.5686	12836	—
USDT/USD B_t (Kraken)	Post-SVB	1m	no-ff	0.1676	0.3881	7701	—
USDT/USD B_t (Kraken)	Post-SVB	5m	all	0.1363	1.7392	2585	—
USDT/USD B_t (Kraken)	Post-SVB	5m	no-ff	0.1549	1.8583	1442	—

is specification-sensitive.

3.2 Stablecoin peg deviations: USDC vs. USDT

The peg dynamics explain the asymmetry between D_t and B_t. On Kraken, USDC trades as low as \$0.87 on March 11 (about $-1{,}300$ bps from \$1), which mechanically creates large positive D_t in BTC/USDC vs. BTC/USD. Over the same window, USDT trades at a premium to \$1, averaging $+20.7$ bps on Kraken and $+23.7$ bps on Coinbase.

Figure 4: Panel A: stablecoin spot prices vs. USD. Panel B: peg deviations from $1.00 (bps). USDC falls sharply while USDT trades at a premium.

3.3 Cross-exchange spatial arbitrage

We also examine the same pair across exchanges by computing log-price differentials for BTC/USDT (Binance vs. Kraken; Coinbase vs. Kraken) and BTC/USD (Coinbase vs. Kraken). The BTC/USDT cross-exchange channel is dispersion in the USDT numeraire; a USD-based arbitrage would require an additional conversion leg unless USDT/USD is identical across venues.

Figure 5: Cross-exchange basis for BTC/USDT (Binance vs. Kraken, Coinbase vs. Kraken) and BTC/USD (Coinbase vs. Kraken). The fiat-to-fiat channel (Panel C) shows persistent positive basis post-crisis.

In calm periods, Binance–Kraken BTC/USDT deviations are essentially zero (mean 0.007 bps). In contrast, Coinbase–Kraken BTC/USD shows a persistent positive mean (+2.18 bps) over the full window, consistent with a structural premium on the main U.S. fiat gateway. During the crisis, all cross-exchange price differentials widen.

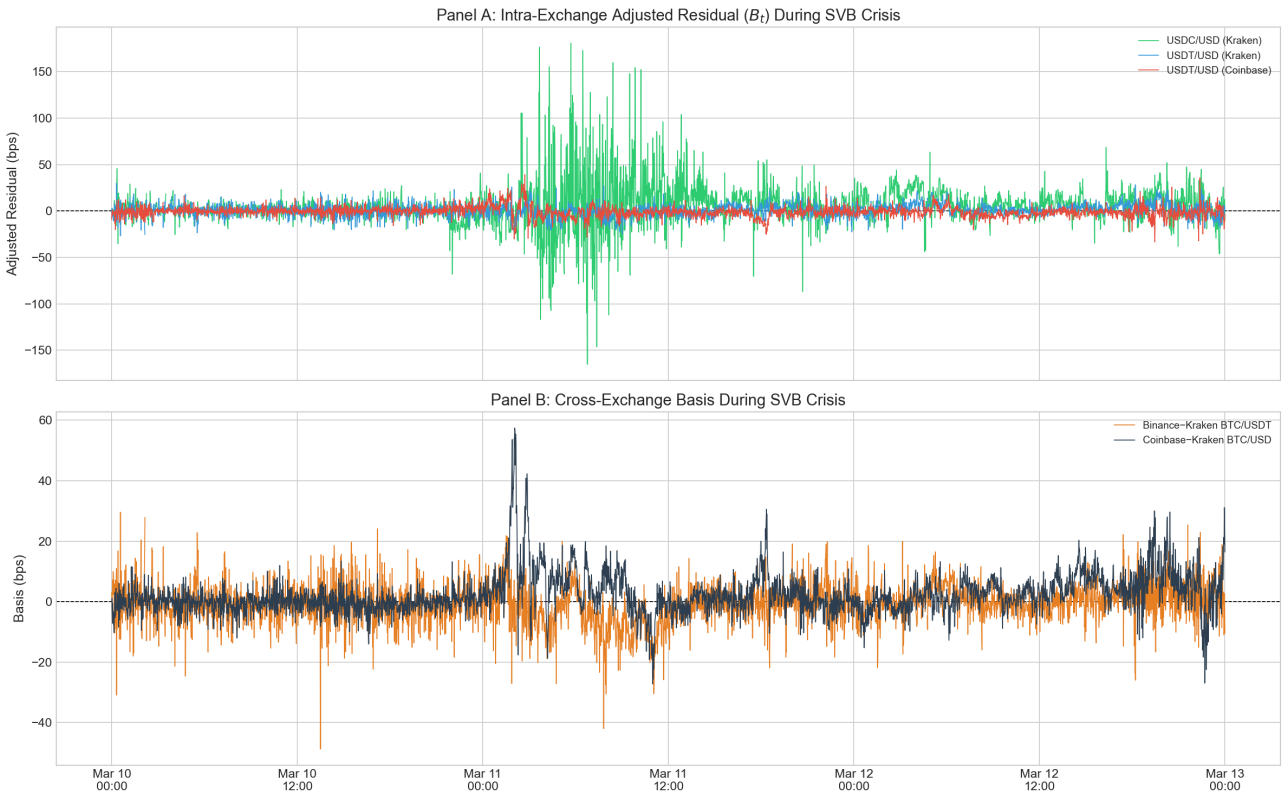

Figure 6: Zoom on March 10–13. Panel A: intra-exchange adjusted residual (B_t). Panel B: cross-exchange basis. Both widen together, indicating system-wide stress rather than a single-venue issue.

3.4 Liquidity, volatility, and explanatory regressions

Using the intra-minute range proxy, trading-friction proxies widen broadly during the crisis, with larger deterioration in stablecoin-quoted pairs.

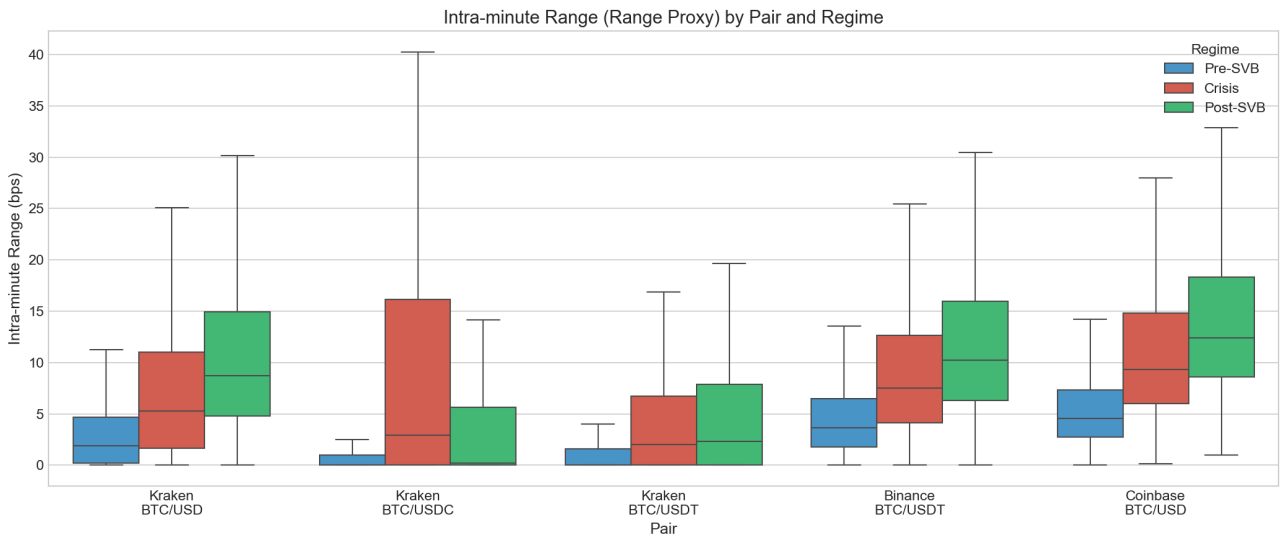

Figure 7: Intra-minute range (range proxy) by pair and regime. Stablecoin pairs deteriorate more during the crisis.

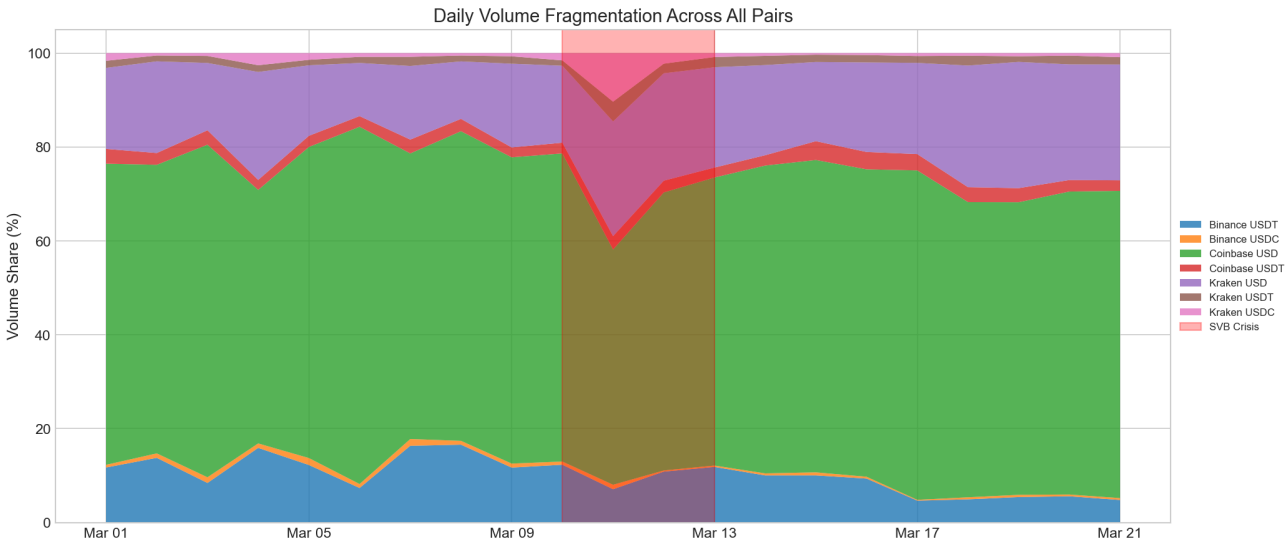

Figure 8: Daily volume shares across exchange–pair combinations, showing how trading reallocates during the SVB window.

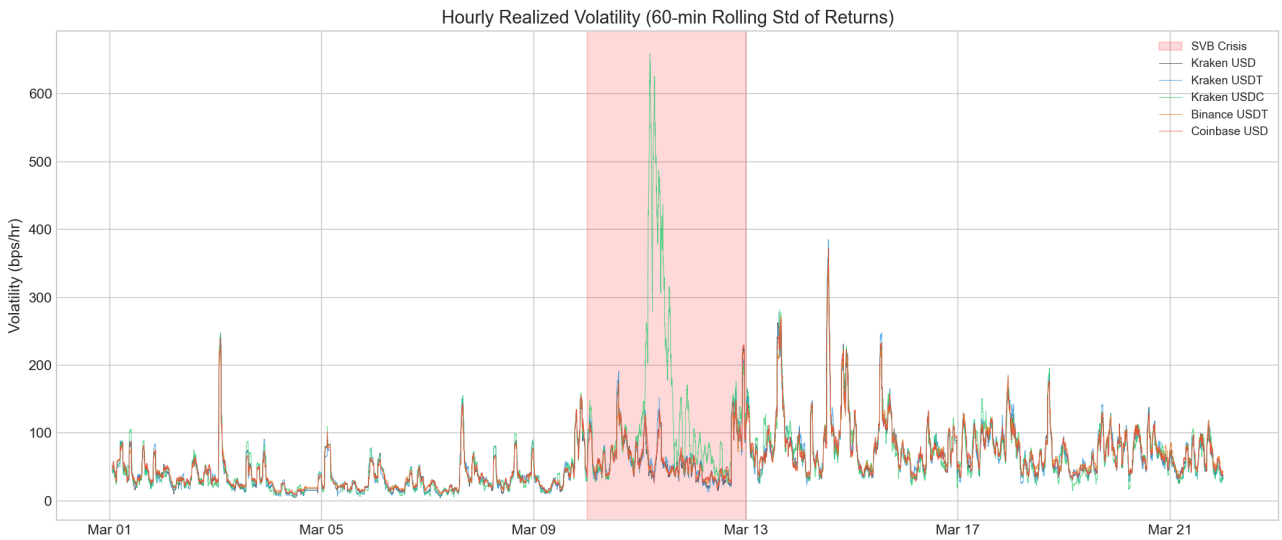

Figure 9: Hourly realized volatility (60-minute rolling standard deviation of returns) across BTC pairs. Volatility spikes during the SVB crisis.

Table 3 shows that the intra-minute range proxy is positively correlated with realized volatility across pairs and regimes. For the regression channels, Corr(Range, RV60m) is 0.5346 overall (0.5510 in crisis) for Kraken BTC/USDC and 0.4560 overall (0.3735 in crisis) for Kraken BTC/USDT. Therefore, the range coefficient in regressions should be interpreted as an association that may reflect both frictions and volatility co-movement, not a clean causal liquidity effect.

We run separate HAC (Newey–West, 60 lags) regressions for the USDC and USDT adjusted residuals (B_t) on Kraken:

Table 3: Correlation Between Intra-minute Range Proxy and 60-minute Realized Volatility

Pair	Regime	Corr(Range, RV60m)	N
Kraken BTC/USD	Overall	0.5852	26489
Kraken BTC/USD	Pre-SVB	0.4922	10093
Kraken BTC/USD	Crisis	0.4730	4249
Kraken BTC/USD	Post-SVB	0.5348	12146
Kraken BTC/USDC	Overall	0.5346	26489
Kraken BTC/USDC	Pre-SVB	0.3017	10093
Kraken BTC/USDC	Crisis	0.5510	4249
Kraken BTC/USDC	Post-SVB	0.3985	12146
Kraken BTC/USDT	Overall	0.4560	26489
Kraken BTC/USDT	Pre-SVB	0.3626	10093
Kraken BTC/USDT	Crisis	0.3735	4249
Kraken BTC/USDT	Post-SVB	0.4433	12146
Binance BTC/USDT	Overall	0.6146	26489
Binance BTC/USDT	Pre-SVB	0.5546	10093
Binance BTC/USDT	Crisis	0.5378	4249
Binance BTC/USDT	Post-SVB	0.5560	12146
Coinbase BTC/USD	Overall	0.6359	26489
Coinbase BTC/USD	Pre-SVB	0.5630	10093
Coinbase BTC/USD	Crisis	0.5164	4249
Coinbase BTC/USD	Post-SVB	0.5736	12146

	USDC Adjusted Residual (B_t)		**USDT Adjusted Residual (B_t)**	
Variable	**Coef**	**P> |z|**	**Coef**	**P> |z|**
Intercept	-0.079	0.833	-0.656	0.000^{***}
Crisis Dummy	4.295	0.000^{***}	1.655	0.000^{***}
Realized Volatility (60m)	0.004	0.928	-0.029	0.179
Intra-minute Range Proxy	0.048	0.005^{**}	-0.013	0.082
R^2	0.041		0.012	

HAC standard errors (60 lags). Dependent variable: adjusted residual B_t (bps).

Table 4: Explanatory regression for USDC and USDT adjusted residuals (Kraken)

The crisis dummy is large and highly significant for both stablecoins, but much larger for USDC (+4.30 bps) than USDT (+1.65 bps). The intra-minute range proxy is positively associated with USDC deviations ($\beta = 0.048$, $p = 0.005$) and only marginally associated for USDT ($p = 0.082$). Given endogeneity and the strong range-volatility co-movement, these coefficients are descriptive associations rather than causal effects.

3.5 Information flow and price discovery

Our main price-discovery evidence uses levels-based cointegration and VECM estimates on the no-forward-fill sample.

Table 5 shows that Kraken BTC/USD vs. BTC/USDT rejects no cointegration (rank = 1) under the BIC-selected lag and remains rank = 1 at $k_\Delta \pm 1$. For Kraken BTC/USD vs. BTC/USDC, the

Table 5: Johansen Cointegration Tests on Log Price Levels (Primary Kraken Channels, No-FF Sample)

Channel	N (No-FF)	VAR p	VECM k_diff	Trace r=0	Crit95 r=0	Reject r=0	Rank Used	Rank (k-1)	Rank (k+1)
Kraken BTC/USD vs BTC/USDC	15710	3	2	7.7052	15.4943	False	0	0	0
Kraken BTC/USD vs BTC/USDT	20415	9	8	27.0257	15.4943	True	1	1	1

Table 6: VECM Adjustment Metrics for Primary Kraken Channels (No-FF Sample)

Channel	Rank	k_delta	alpha_1	alpha_2	Leader by $\lvert\alpha\rvert$
Kraken BTC/USD vs BTC/USDC	0	2	NaN	NaN	undetermined_no_cointegration
Kraken BTC/USD vs BTC/USDT	1	8	0.0079	0.0118	Kraken BTC/USD

conservative no-FF specification does *not* reject no cointegration (rank $= 0$), and this remains rank $= 0$ at $k_\Delta \pm 1$. Therefore, we do not treat "USDC follows fiat" as a primary VECM-identified discovery result in this specification.

For the cointegrated USDT channel, the VECM adjustment coefficients satisfy $|\alpha_{\text{BTC/USD}}| < |\alpha_{\text{BTC/USDT}}|$ (0.0079 vs. 0.0118), so BTC/USD adjusts less and BTC/USDT adjusts more to restore equilibrium. The inferred leader direction is stable at $k_\Delta \pm 1$. We report GG diagnostics in Appendix Table 9; leader direction is inferred from $|\alpha|$.

We keep Granger causality as secondary evidence and apply BH/FDR correction across the full test family:

In the Kraken USDC channel, the directional Granger asymmetry (BTC/USD $\rightarrow$ BTC/USDC significant; reverse not significant after FDR) is consistent with fiat-to-stablecoin adjustment, but we treat it as supportive because 1-minute asynchrony and large-N sensitivity can overstate predictive causality. We remove orthogonalized IRFs from the main narrative because they are ordering-dependent.

3.6 Arbitrage after fees

We evaluate arbitrage with explicit trade construction and costs. For intra-exchange stablecoin parity channels on Kraken (BTC/USDC vs. BTC/USD and BTC/USDT vs. BTC/USD), we use a 3-leg triangular trade (BTC/SC, SC/USD, BTC/USD), so $n_{\text{legs}} = 3$. For cross-exchange channels (Binance–Kraken BTC/USDT and Coinbase–Kraken BTC/USD), we use a 2-leg pre-funded buy/sell setup, so $n_{\text{legs}} = 2$.

Table 7: Granger Causality Tests with BH/FDR Correction (Secondary Evidence)

Channel	F-stat	p-value	Sig($p < 0.05$)	q-value (BH/FDR)	Sig($q <$ 0
BTC/USDC $\rightarrow$ BTC/USD (Kraken)	1.512000	0.127898	No	0.127898	No
BTC/USD $\rightarrow$ BTC/USDC (Kraken)	332.178000	0.000000	Yes	0.000000	Yes
BTC/USDT $\rightarrow$ BTC/USD (Kraken)	3.825000	0.000034	Yes	0.000046	Yes
BTC/USD $\rightarrow$ BTC/USDT (Kraken)	385.271000	0.000000	Yes	0.000000	Yes
Kraken USDT $\rightarrow$ Binance USDT	7.265000	0.000000	Yes	0.000000	Yes
Binance USDT $\rightarrow$ Kraken USDT	476.065000	0.000000	Yes	0.000000	Yes
Kraken USD $\rightarrow$ Coinbase USD	3.709000	0.000055	Yes	0.000063	Yes
Coinbase USD $\rightarrow$ Kraken USD	130.133000	0.000000	Yes	0.000000	Yes

Table 8: Arbitrage Profitability Bounds by Channel and Regime (5 bps per taker leg; 2-leg pre-funded cross-exchange and 3-leg intra-exchange stablecoin channels)

Channel	Regime	Cost Variant	MeanAbs (bps)	%Profitable	AvgNetCond (bps)	AvgNetUncond (bps)
Cross BTC/USD (Coinbase-Kraken)	Crisis	Fee-only	4.74	11.32	6.36	0.72
Cross BTC/USD (Coinbase-Kraken)	Crisis	Fee + slippage	4.74	2.29	6.37	0.15
Cross BTC/USD (Coinbase-Kraken)	Post-SVB	Fee-only	5.98	19.36	7.38	1.43
Cross BTC/USD (Coinbase-Kraken)	Post-SVB	Fee + slippage	5.98	6.66	4.67	0.31
Cross BTC/USD (Coinbase-Kraken)	Pre-SVB	Fee-only	1.76	0.19	1.61	0.00
Cross BTC/USD (Coinbase-Kraken)	Pre-SVB	Fee + slippage	1.76	0.00	0.00	0.00
Cross BTC/USDT (Binance-Kraken)	Crisis	Fee-only	4.84	11.85	4.12	0.49
Cross BTC/USDT (Binance-Kraken)	Crisis	Fee + slippage	4.84	1.81	2.75	0.05
Cross BTC/USDT (Binance-Kraken)	Post-SVB	Fee-only	5.12	12.57	4.26	0.54
Cross BTC/USDT (Binance-Kraken)	Post-SVB	Fee + slippage	5.12	1.87	3.34	0.06
Cross BTC/USDT (Binance-Kraken)	Pre-SVB	Fee-only	3.13	4.54	3.90	0.18
Cross BTC/USDT (Binance-Kraken)	Pre-SVB	Fee + slippage	3.13	0.98	2.90	0.03
USDC/USD (Kraken, intra)	Crisis	Fee-only	13.37	29.16	16.58	4.83
USDC/USD (Kraken, intra)	Crisis	Fee + slippage	13.37	6.72	9.56	0.64
USDC/USD (Kraken, intra)	Post-SVB	Fee-only	6.68	9.38	5.98	0.56
USDC/USD (Kraken, intra)	Post-SVB	Fee + slippage	6.68	1.78	4.22	0.08
USDC/USD (Kraken, intra)	Pre-SVB	Fee-only	3.53	1.50	4.11	0.06
USDC/USD (Kraken, intra)	Pre-SVB	Fee + slippage	3.53	0.21	3.04	0.01
USDT/USD (Kraken, intra)	Crisis	Fee-only	4.94	3.76	3.79	0.14
USDT/USD (Kraken, intra)	Crisis	Fee + slippage	4.94	0.37	2.51	0.01
USDT/USD (Kraken, intra)	Post-SVB	Fee-only	4.92	3.44	4.41	0.15
USDT/USD (Kraken, intra)	Post-SVB	Fee + slippage	4.92	0.40	2.70	0.01
USDT/USD (Kraken, intra)	Pre-SVB	Fee-only	3.02	1.07	3.90	0.04
USDT/USD (Kraken, intra)	Pre-SVB	Fee + slippage	3.02	0.19	3.41	0.01

We report two bounds for each channel/regime:

$$\text{cost}_t^{\text{fee}} = n_{\text{legs}} \cdot f, \tag{11}$$

$$\text{cost}_t^{\text{fee+slip}} = n_{\text{legs}} \cdot f + \sum_{\ell=1}^{n_{\text{legs}}} \tfrac{1}{2}\,\text{RangeProxy}_{\ell,t}, \tag{12}$$

$$\text{net}_t = (|b_t| - \text{cost}_t)_+, \tag{13}$$

with $f = 5$ bps per taker trade (2-leg = 10 bps, 3-leg = 15 bps). The per-leg range term is a *range-based slippage proxy*, not a quoted spread; it is conservative and noisy. Here b_t denotes the relevant mispricing series (B_t for intra-exchange stablecoin channels; cross-exchange log-price differential for venue channels).

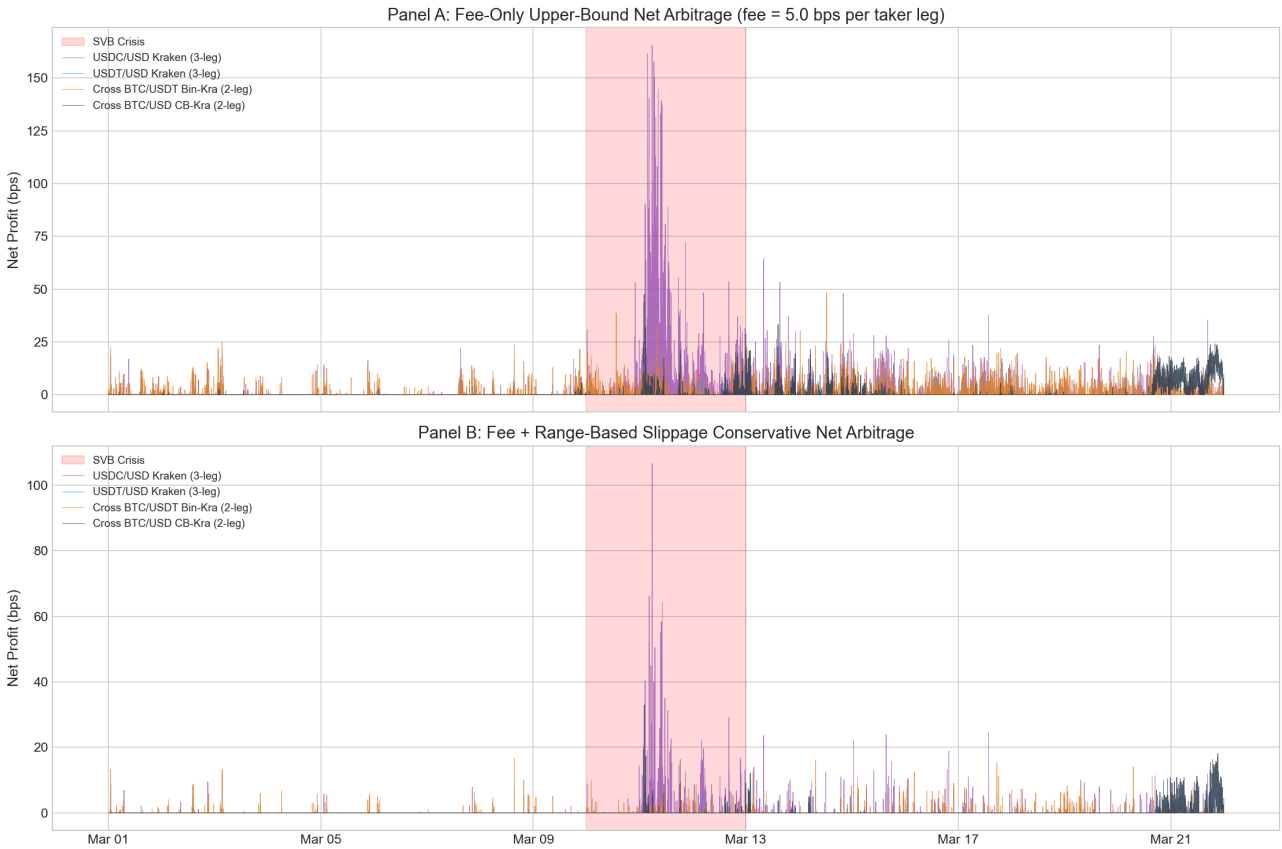

Figure 10: Net arbitrage opportunity across four channels under two bounds. Panel A: fee-only upper bound. Panel B: fee + range-based slippage conservative bound.

The fee-only specification is an optimistic upper bound; adding the range-based slippage term reduces both profitability frequency and net capture. In the USDC/USD Kraken crisis row, profitability falls from 29.16% (fee-only) to 6.72% (fee+slippage), and unconditional net falls from 4.83 bps to 0.64 bps. In the Coinbase–Kraken BTC/USD crisis row, profitability falls from 11.32% to 2.29% (unconditional net 0.72 bps to 0.15 bps). For cross-exchange channels, realizability requires pre-funded inventory on both venues and ignores transfer delays; results also abstract from exchange-specific fee tiers and execution queue dynamics.

4 Regulatory Interpretation

The SVB episode shows that stablecoin "pegs" can fail because of the banking layer (custody, access to reserves, redemption uncertainty), not because of crypto price mechanics. The GENIUS Act targets exactly this channel by imposing reserve standards, transparency, and oversight, which should reduce the probability and severity of a confidence-driven de-peg. In our setting, that would directly reduce the crisis-driven USDC unadjusted dispersion blowout (mean +319.97 bps on Kraken) and also compress the adjusted residual (mean +5.27 bps), tightening cross-currency parity.

Our results also highlight that stablecoins are not interchangeable under stress: USDC behaves like a solvency/custody risk event, while USDT behaves like a demand premium event. A uniform regulatory framework should narrow these differences by reducing uncertainty about reserve quality and redemption.

Table 9: Gonzalo–Granger Diagnostics (Appendix; supplementary; not shares and may be outside [0,1])

Channel	Rank	k_delta	GG diag mkt1	GG diag mkt2	GG note
Kraken BTC/USD vs BTC/USDC	0	2	NaN	NaN	no_rank1_cointegration
Kraken BTC/USD vs BTC/USDT	1	8	2.9813	-1.9813	gg_non_convex_share

Finally, the persistence of fiat cross-exchange deviations (Coinbase–Kraken BTC/USD) suggests that even regulated fiat rails do not fully eliminate fragmentation. More reliable stablecoin settlement (with credible reserves) could allow market makers to move capital and inventory more efficiently across venues, reducing these spatial frictions.

5 Conclusion

Using 1-minute data around the SVB crisis (March 1–21, 2023), we quantify how stablecoin stress affects cross-currency parity, liquidity, and price discovery in BTC spot markets. The main findings are:

1. The USDC de-peg produces large unadjusted dispersion D_t, but the adjusted residual B_t remains strongly mean-reverting with very short half-life (USDC crisis: 0.61 minutes at 1-minute sampling; robustness 0.54–1.10 minutes across 1m/5m and all/no-FF specs).

2. The crisis generates a much larger structural break for USDC than USDT (+4.30 bps vs. +1.65 bps), and the intra-minute range proxy is positively associated with larger USDC deviations.

3. In levels-based discovery tests on the primary Kraken channels, BTC/USD vs. BTC/USDT is cointegrated and BTC/USD adjusts less to disequilibrium, while BTC/USD vs. BTC/USDC does not reject no-cointegration under the conservative no-FF BIC specification; USDC-following-fiat is therefore treated as supportive Granger evidence, not a primary VECM-identified result.

4. Arbitrage profitability is highly model-dependent: fee-only calculations provide an upper bound, while adding a range-based slippage proxy materially reduces net opportunities, especially in stablecoin stress windows.

These patterns connect directly to stablecoin regulation: when redemption and reserves are credible, stablecoin-specific risk premia shrink, cross-currency basis tightens, and market fragmentation becomes easier to arbitrage away.

Appendix